CS 529
Game Engines Design

Instructor

- Mariojulio Zaldivar
  - mariojulio.alvarez@digipen.edu
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DigiPen INSTITUTE OF TECHNOLOGY

Week 2 Lecture

- ❑Math objects definition
- ❑Transforms

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Vector

APIs for vector operations

Vector definition

- Vector
  - Properties
    - x, y
    - x, y, z
  - Functions
    - Algebraic operations

Accessibility

Members: public or private?

Encapsulation with getters:

```
Vector3D vec;
vec.getX(); // slow… but why?
```

Accessibility

Members: public or private?

Encapsulation with getters:

Vector3D vec;

vec.getX(); // slow… but why?

Returns a copy of the value generated in the function scope

Solution?

Return a pointer then?

Solution?

Return a pointer then?

NO!

Solution?

Return a pointer then?

NO!

Return a reference??

Solution?

Return a pointer then?

NO!

Return a reference??

NO!!

Trade-off time

Make members public when speed is critical

-> But ask if you really want to allow developers to modify the members directly.

Make members private to ensure safety of the internal logic of your object

-> But ask at what cost? Returning values from a function might be too costly for some video games.

Matrix

APIs for matrix operations

Matrix

- Matrix
  - Properties
    - 3 x 3 array
    - 4 x 4 array
    - Make it simple, no need for fancy data structures here :)
  - Functions
    - Algebraic operations between matrices
    - Algebraic operations with your class Vector

Functions examples

Identity():

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Multiplication():

$$\begin{bmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ a_7 & a_8 & a_9 \end{bmatrix} \begin{bmatrix} b_1 & b_2 & b_3 \\ b_4 & b_5 & b_6 \\ b_7 & b_8 & b_9 \end{bmatrix} = \begin{bmatrix} c_1 & c_2 & c_3 \\ c_4 & c_5 & c_6 \\ c_7 & c_8 & c_9 \end{bmatrix}$$

Transpose()

…

Stop here!

Multiplication of Matrix with Vector

$$\begin{pmatrix} x0 & 0 & 0 & 0 \\ 0 & y0 & 0 & 0 \\ 0 & 0 & z0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \times \quad (x1, y1, z1)$$

Maths in action: What is the problem here?

Accessibility

Same logic as vector

(and in all APIs to be implemented)

Transform

APIs for objects transforms

Transform

Before coding… what are we coding here?

Spatial transforms

Assume a matrix:

$$\begin{pmatrix} 1 & 0 & 0 & x0 \\ 0 & 1 & 0 & y0 \\ 0 & 0 & 1 & z0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Spatial transforms

Assume a matrix:

$$\begin{pmatrix} 1 & 0 & 0 & x0 \\ 0 & 1 & 0 & y0 \\ 0 & 0 & 1 & z0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Now assume a vector (movement in space)

(5, 0, 0, 1)

Spatial transforms

Multiply the vector by the matrix

$$\begin{pmatrix} 1 & 0 & 0 & x0 \\ 0 & 1 & 0 & y0 \\ 0 & 0 & 1 & z0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{matrix} 5 \\ 0 \\ 0 \\ 1 \end{matrix} = \begin{matrix} x0 + 5 \\ y0 \\ z0 \\ 1 \end{matrix}$$

which represents now a vector in position = (x0+5, y0, z0)

Translation Matrix

$$\begin{pmatrix} 1 & 0 & 0 & x0 \\ 0 & 1 & 0 & y0 \\ 0 & 0 & 1 & z0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Scale Matrix

$$\begin{pmatrix} sx & 0 & 0 & 0 \\ 0 & sy & 0 & 0 \\ 0 & 0 & sz & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Rotation Matrix

3 matrices (for each axis)

$$Rx = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos & \sin & 0 \\ 0 & -\sin & \cos & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$Ry = \begin{pmatrix} \cos & 0 & -\sin & 0 \\ 0 & 1 & 0 & 0 \\ \sin & 0 & \cos & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$Rz = \begin{pmatrix} \cos & -\sin & 0 & 0 \\ \sin & \cos & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

How to transform?

Multiply the position vector:

Translation * Rotation * Scale * vector

How to transform?

Multiply the position vector:

```
if (TransType == LOCAL)
            ModelToWorld = rotate * translate * scale;
if (TransType == WORLD)
            ModelToWorld = translate * rotate * scale;
```